INFO6007

Project Management in IT

Week 02: PM Methodologies and Requirement Gathering

Semester 1, 2026

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Acknowledgement of Country

I would like to acknowledge the Traditional Owners of Australia and recognise their continuing connection to land, water and culture. I am currently on the land of the Gadigal people of the Eora nation and pay my respects to their Elders, past, present and emerging.

I further acknowledge the Traditional Owners of the country on which you are on and pay respects to their Elders, past, present and future.

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Copyright Warning

Agenda

- Introduction
- Project Management Methodologies
- Requirement Gathering
- Project Initiation

Project Lifecycle

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What is IT Project Lifecycle

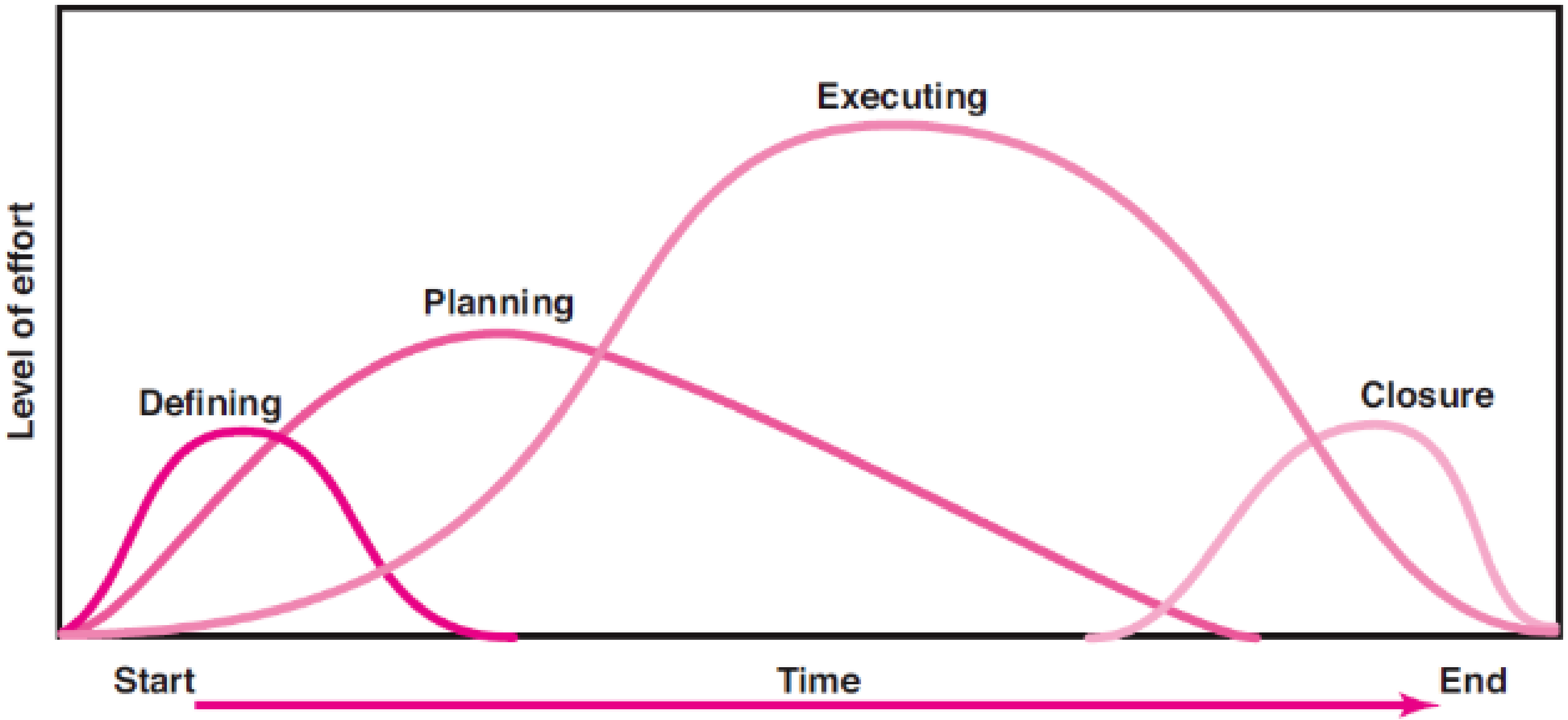
- The **Project Lifecycle** is the structured sequence of phases a project goes through from **start to finish**.

- It provides a clear road map for managing work effectively, reducing risks, and ensuring success.

Project Lifecycle – The different phases

- Initiation
  - Purpose: Define the project at a high level and evaluate its feasibility.
  - Output: Project is approved and formally initiated.
- Planning
  - Purpose: Build a comprehensive plan that guides the team.
  - Output: Approved Project Management Plan
- Execution
  - Purpose: Perform the actual work according to the plan.
  - Output: Project deliverables are developed and reviewed
- Monitoring & Controlling
  - Purpose: Track performance and make adjustments as needed.
  - Output: Status reports, updates, and course corrections
- Closure
  - Purpose: Wrap up the project, deliver results, and reflect.
  - Output: Final project report, lessons learned, client sign-off

Project Management Framework - Project Lifecycle



Defining
1. Goals
2. Specifications
3. Tasks
4. Responsibilities

Planning
1. Schedules
2. Budgets
3. Resources
4. Risks
5. Staffing

Executing
1. Status reports
2. Changes
3. Quality
4. Forecasts

Closure
1. Train customer
2. Transfer documents
3. Release resources
4. Evaluation
5. Lessons learned

Figure 2.1: Project Life Cycle

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Project Management Methodologies

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Introduction

- Project Management Methodologies are structured approaches or systems of practices used to plan, execute, and manage projects efficiently.

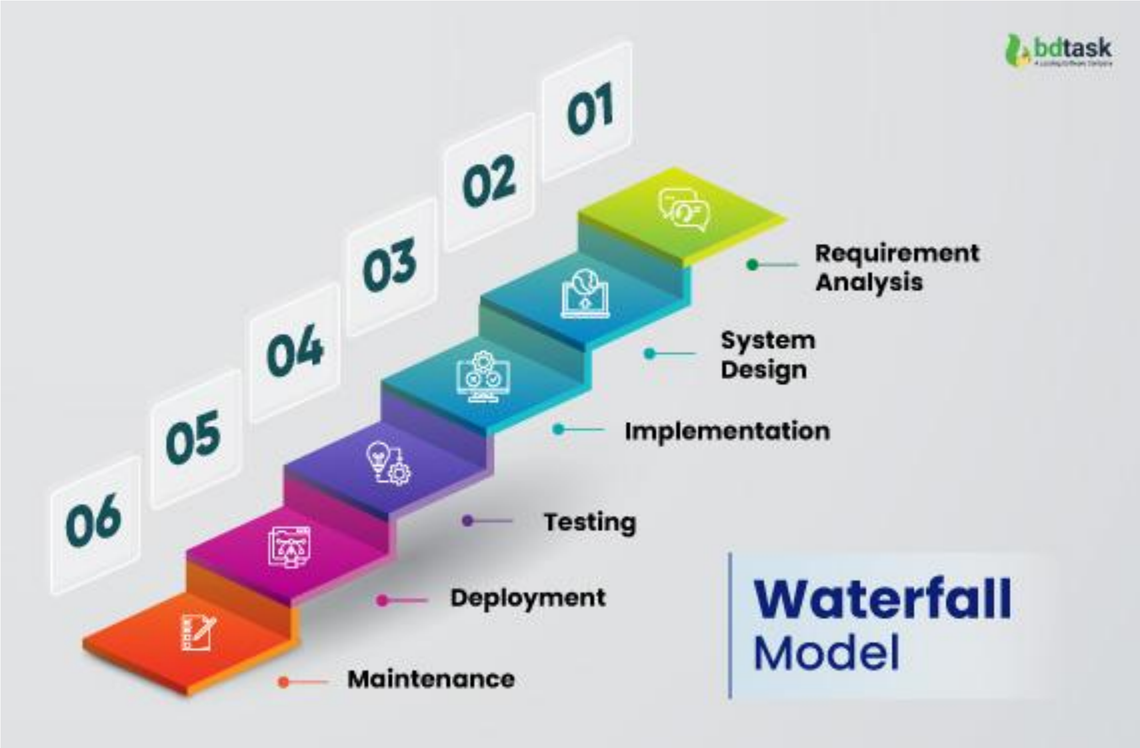
- They provide a **repeatable framework** that guides teams on how to deliver work successfully from start to finish.

- Why is it important?
  - Ensure consistency and predictability
  - Improve team collaboration
  - Manage risks, time, and costs
  - Align work with business goals

Waterfall Project Management

- The **Waterfall Model** is a **linear and sequential** project management methodology where each phase must be **completed before the next begins**.

- It is one of the **earliest and most traditional** approaches, often used in engineering, construction, and structured IT projects.



https://www.bdtask.com/blog/waterfall-model-in-software-engineering

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Waterfall Project Methodology contd.

- Phases of the waterfall model
  - Requirements Gathering - All requirements of the project are collected up-front and no changes are expected once this phase ends.
  - Systems Design - Create a high level and detailed design of the system. This defines the architecture, components, data models, etc.
  - Implementation (Coding) - Developers write code based on the design documents and the development is done usually in modules or components.
  - Testing – System is tested for defects, bugs, and performance. This typically include unit testing, integration testing, and system testing.
  - Deployment - Product is delivered to users or moved to a production environment.
  - Maintenance – Fix issues, update software, and provide support after deployment.

Waterfall Project Methodology contd.

- Advantages
  - Simple to understand and use
  - Clear milestones and documentation
  - Easier to manage for projects with fixed scope and stable requirements
  - Ideal for government, construction, or compliance-heavy projects
- Disadvantages
  - Not flexible to changes once development begins
  - Difficult to go back to a previous phase
  - Delayed testing can uncover late-stage problems
  - Poor fit for dynamic or evolving requirements

Example #1 – Waterfall Methodology

- A mid-sized manufacturing company with 500 employees decides to implement a **new Human Resource Management System (HRMS)** to replace its outdated, paper-based system. The new system will include modules for employee records, payroll processing, leave management, and performance tracking.
- The **senior leadership mandates a strict timeline** of 8 months, and the scope and budget are **clearly defined upfront**. The vendor chosen for development has requested a detailed **requirement specification document** before starting any work. The company has no prior experience with agile development and prefers a structured, linear delivery process.

- Why is Waterfall methodology chosen here?
  - Requirements are well understood and unlikely to change
  - Project success depends on complete documentation
  - Stakeholders prefer formal approvals at each phase
  - Development is outsourced and requires clear handoffs

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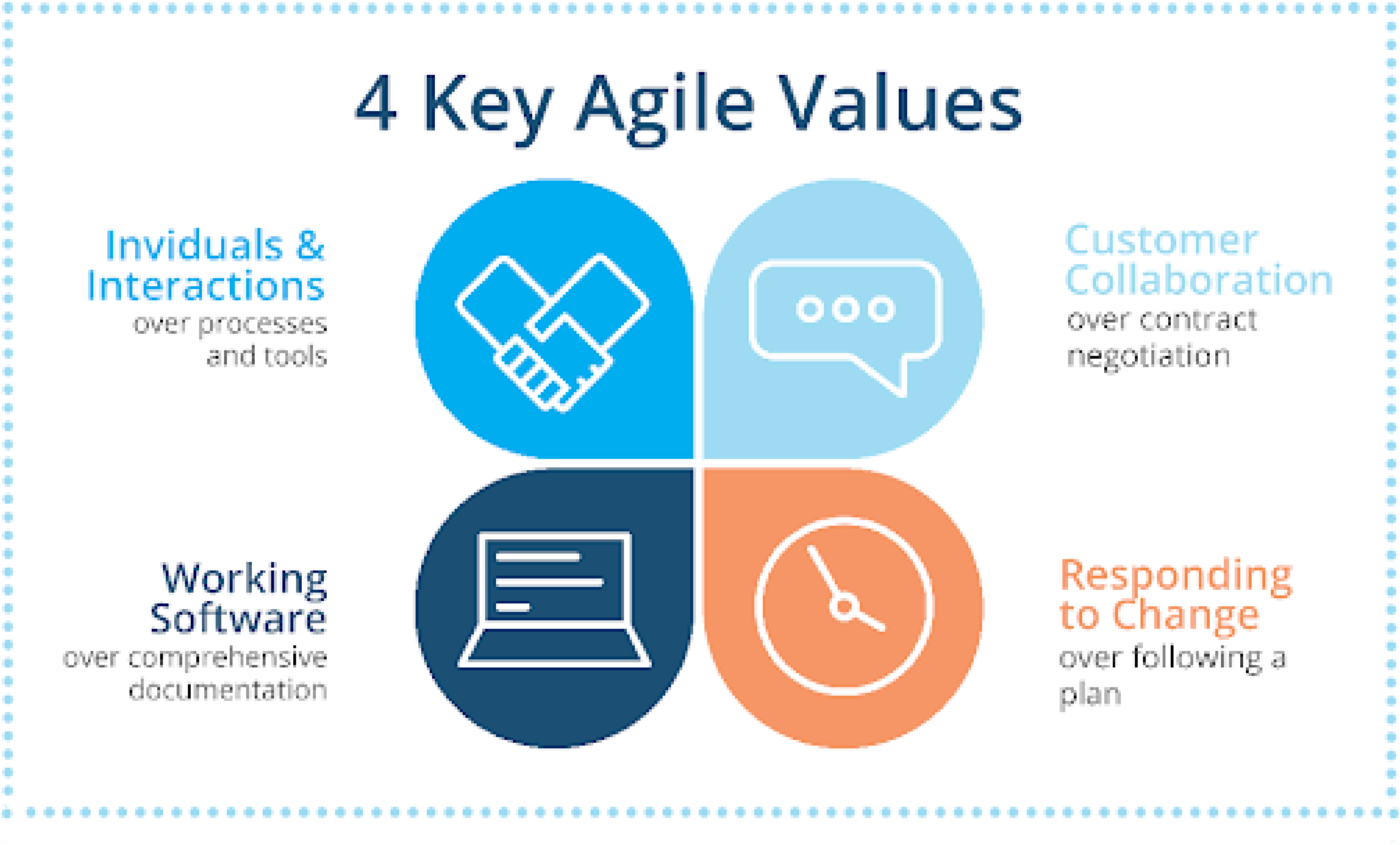
Agile Project Methodology

- Agile is a flexible, iterative approach to project management, primarily used in software development and dynamic project environments.

- Unlike rigid, linear models (like Waterfall), Agile breaks the project into small, manageable increments called sprints or iterations.

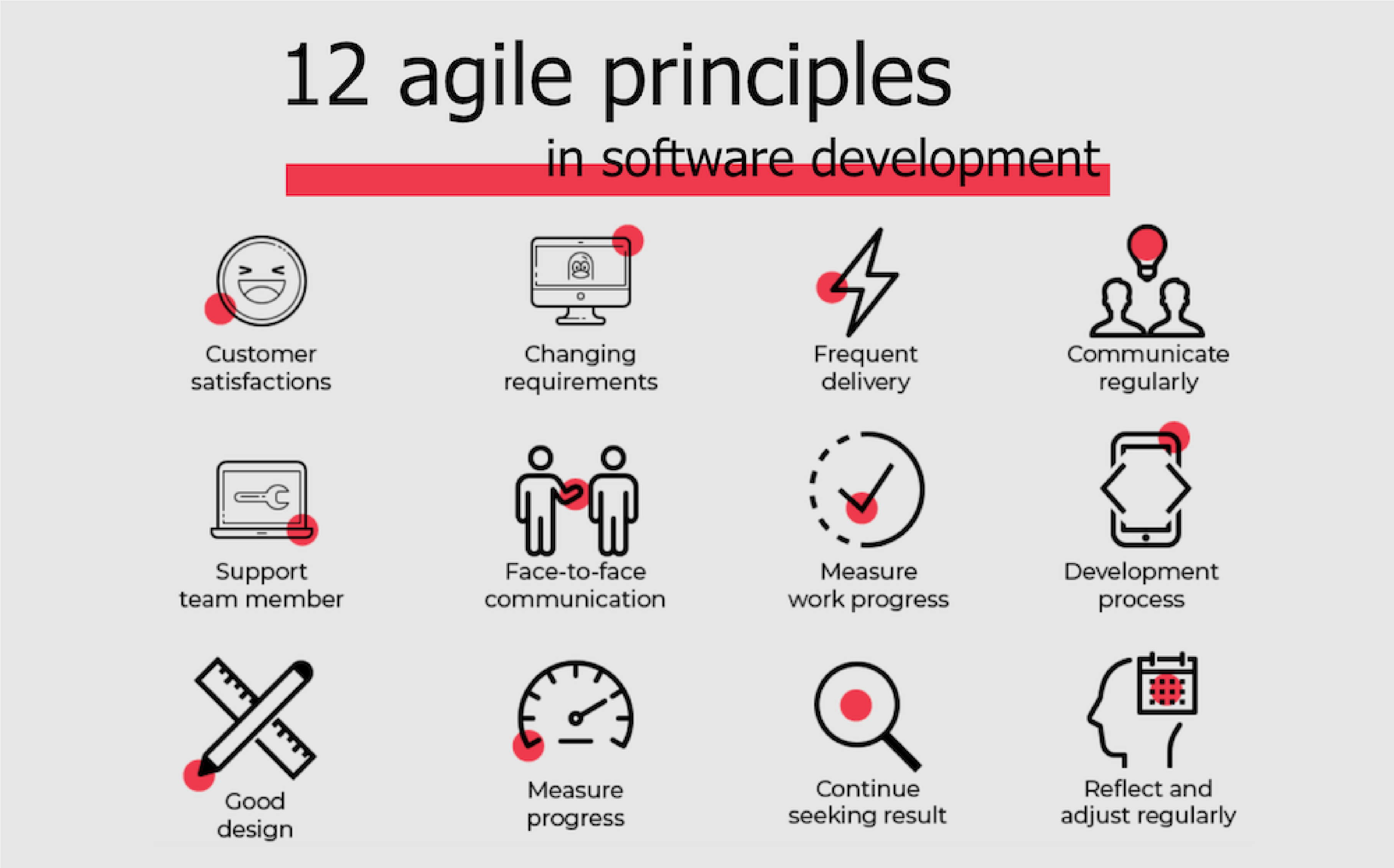
- Remember, it is not a management technique but a mindset!

Agile Project Methodology contd.



(What Is Agile Software Development? Agile Principles and Values, 2020)

Agile Project Methodology contd.



(What Is Agile Software Development? Agile Principles and Values, 2020)

Scrum Methodology

- Scrum is an Agile framework that delivers work in time-boxed iterations called Sprints (typically 1 to 4 weeks), focusing on empirical process control, team collaboration, and continuous feedback.

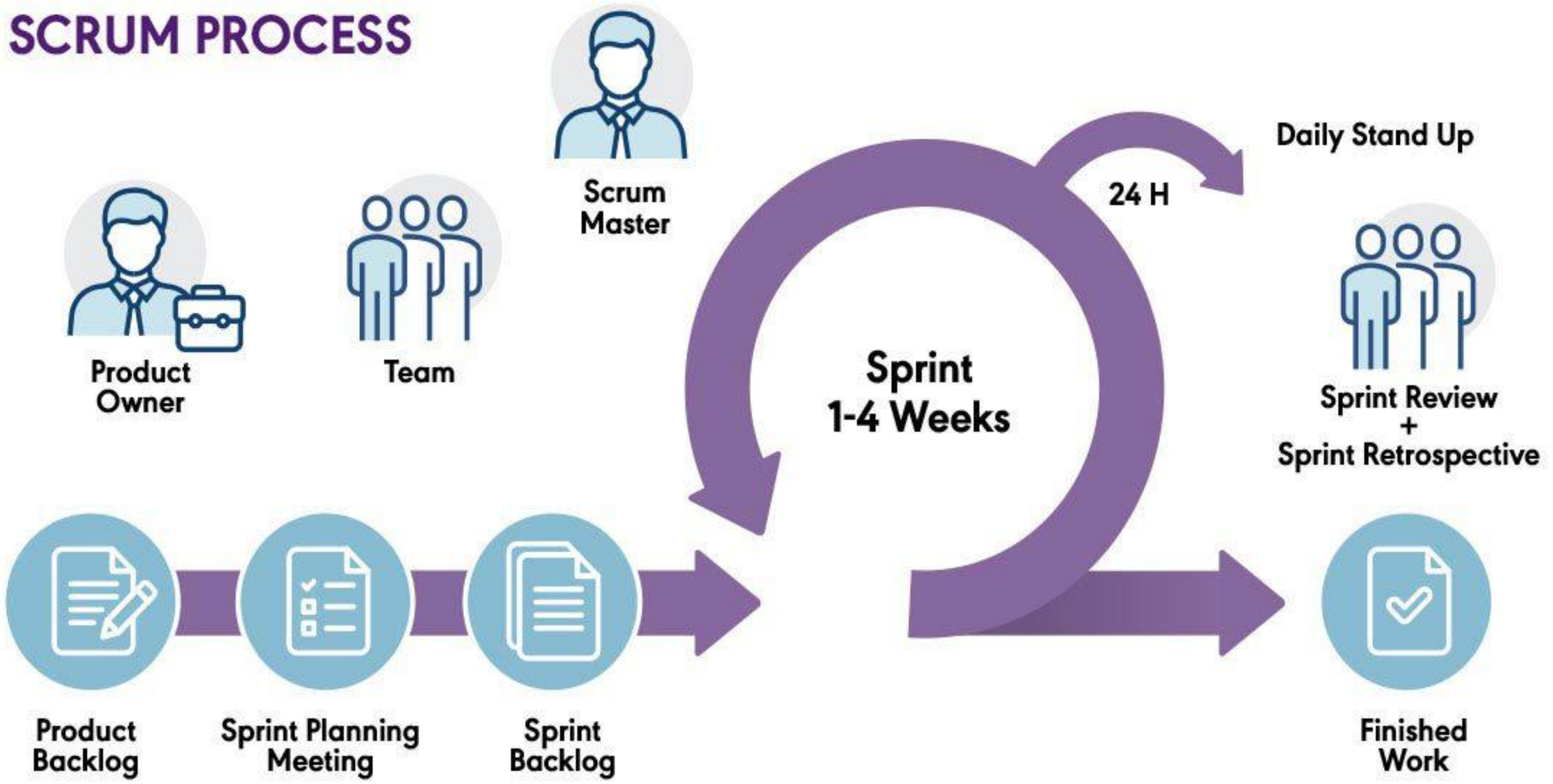
- Key components of Scrum
  - Sprint - Time-boxed development cycle (ranges from 1 to 4 weeks usually)
  - Product Backlog - List of all desired features (maintained by Product Owner)
  - Sprint Backlog - List of tasks to be completed in the current Sprint
  - Increment - The potentially shippable product at the end of the Sprint

Scrum methodology contd.

- Scrum Roles
  - Product Owner - Defines the features, manages backlog, prioritizes work
  - Scrum Master - Coaches the team, facilitates Scrum events, removes blockers
  - Scrum Team - Cross-functional members who do the actual work

- Scrum Events
  - Sprint Planning - Define what will be done this Sprint
  - Daily Stand up - 15-min quick check-in usually over a coffee
  - Sprint Review - Showcase work to stakeholders
  - Sprint Retrospective - Reflect and improve

Scrum methodology contd.



(What is Scrum? An overview of Scrum and The Agile Journey, 2024)

Example #2 – Scrum Methodology

- A university IT department receives funding to develop a **mobile app** that improves **student engagement** by integrating features such as event notifications, campus maps, peer messaging, and feedback surveys. The users are current students and university staff.
- Requirements are **evolving** as different departments request features. The Product Owner emphasizes **incremental delivery** so students can test and give feedback regularly. The app must go live in 4 months, with core features delivered first.

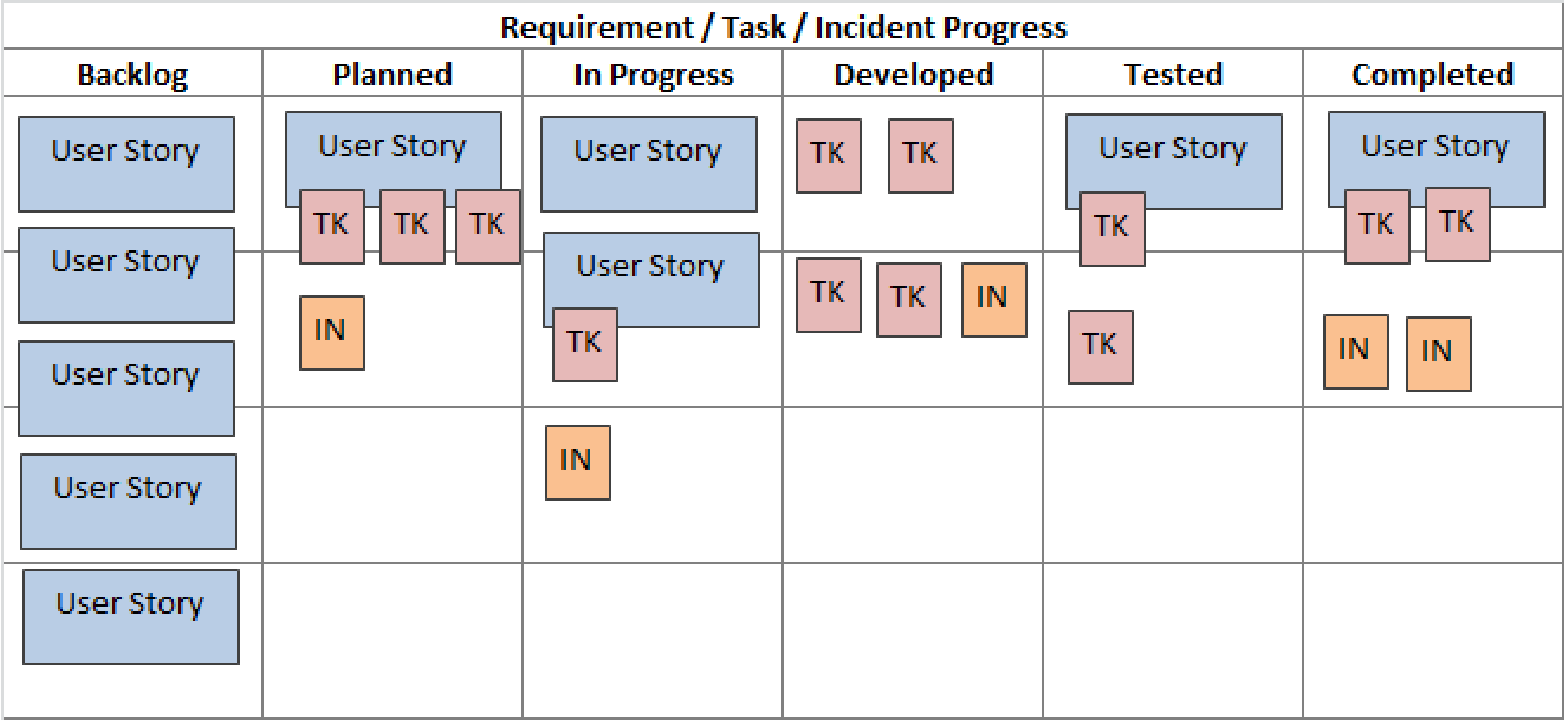
- Why was Scrum chosen?
  - Requirements are **not fixed upfront**
  - Prioritization changes based on student feedback
  - Frequent **demos and stakeholder input** needed
  - Team is small and cross-functional
  - Value delivery is emphasized over documentation

Kanban methodology

- Kanban is a visual workflow management method that focuses on continuous delivery, limiting work in progress (WIP), and workflow transparency.

- Key elements of Kanban
  - Kanban Board - Visual board that tracks tasks through stages (To Do → Doing → Done)
  - Cards/User stories - Individual tasks or work items
  - WIP Limits - Restriction on number of items in a column to prevent overload
  - Cycle Time - Time taken for a task to go from start to finish

Kanban methodology contd.



(Kanban Methodology Guide & Explainer, n.d.)

Example #3 – Kanban Methodology

- A growing software company wants to improve its **customer support ticket handling process**. Currently, tickets pile up in a backlog and often get delayed because the support team struggles to prioritize tasks and balance workload.
- The support team consists of 5 agents who handle bug reports, feature requests, and user questions. The company wants a **visual system** to track ticket progress, limit work in progress (WIP), and continuously improve efficiency.
- There is no strict deadline for resolving tickets, but the company aims to reduce response times and increase customer satisfaction.

- Why Kanban was chosen?
  - Continuous flow of work with no fixed-length iterations
  - Flexibility to handle incoming tickets anytime
  - Visual management of tasks to identify bottlenecks
  - Limits on work in progress to avoid overloading agents
  - Emphasis on continuous improvement (Kaizen)

Scrum vs Kanban

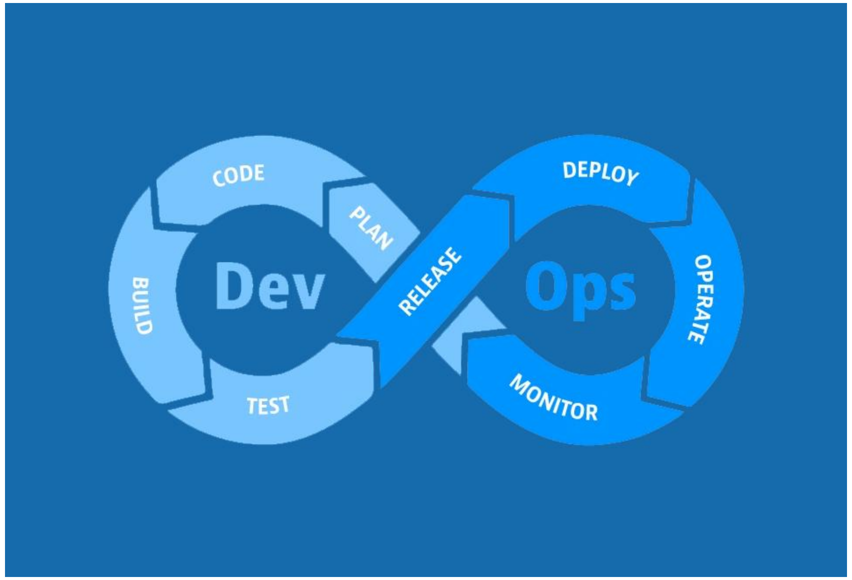
Feature	Scrum	Kanban
Structure	Time boxed sprints	Continuous flow
Roles	Defined (PO, SM, Team)	No specific roles required
Workload	Planned per sprint	Pulled as capacity allows
Change Handling	Limited during Sprint	Change is allowed anytime
Visual Management	Optional board	Mandatory Kanban board
Best For	Teams needing structure	Teams needing flexibility

DevOps

- A set of practices that integrates software development and IT operations to shorten the development lifecycle, increase deployment frequency, and deliver high-quality software continuously.

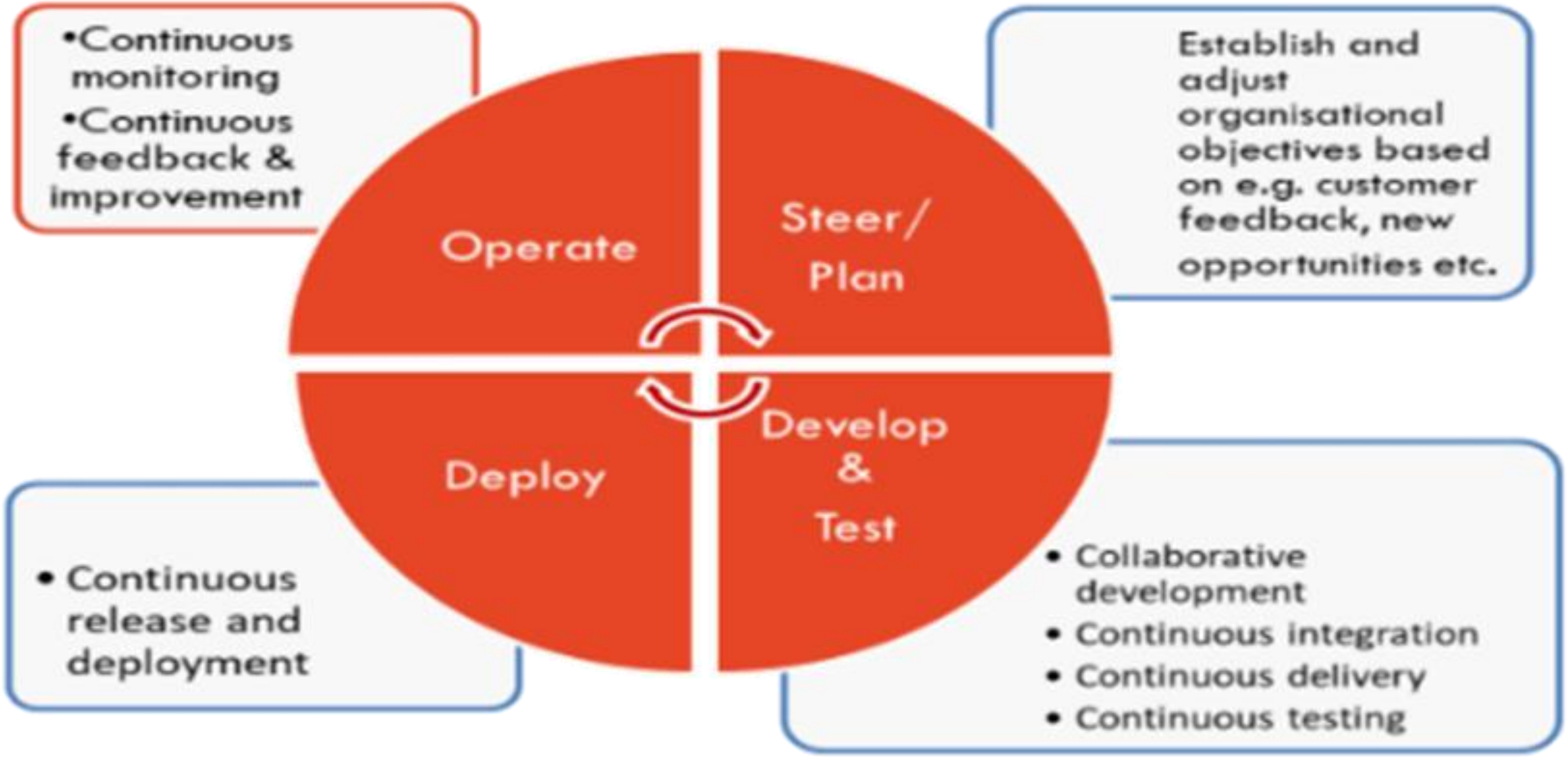
- DevOps = Development + Operations

- The main goal for DevOps are:
  - Faster delivery cycles (CI/CD)
  - Improved collaboration between teams
  - Automated testing, deployment, and monitoring
  - Infrastructure as Code (IaC) for scalability and reliability



T2informatik, (2024)

Iterative phases of the DevOps cycle



DevOps Culture Scales Through Automation

DevOps is a **culture**,
but culture scales only through **automation and tooling**.

- Without tools, DevOps cannot scale.
- With tools, DevOps becomes continuous delivery.

Culture defines behavior.
Automation enforces behavior at scale.

DevOps Toolchain Overview

DevOps Phase	Example Tools	Purpose of the Tools
Plan	Jira, Confluence, XMind, draw.io	Backlog management, sprint planning, documentation, requirement modeling
Code / Develop	Git, GitHub, GitLab, Bitbucket	Version control, collaboration, code review, branch management
Build	Maven, Gradle, Webpack, Grunt	Dependency management, compilation, packaging, artifact creation
Continuous Integration (CI)	Jenkins, GitHub Actions, CircleCI	Automated builds, automated testing, integration validation on commit
Test	JUnit, NUnit, Pytest, Selenium, JMeter, BlazeMeter	Unit testing, UI testing, performance & load testing
Release & Deploy	Docker, Kubernetes, AWS, Azure, Artifact Registry	Containerization, orchestration, scalable cloud deployment
Continuous Monitoring	Prometheus, Grafana, New Relic, Sumo Logic, Slack	Metrics collection, observability, incident alerting, feedback loop

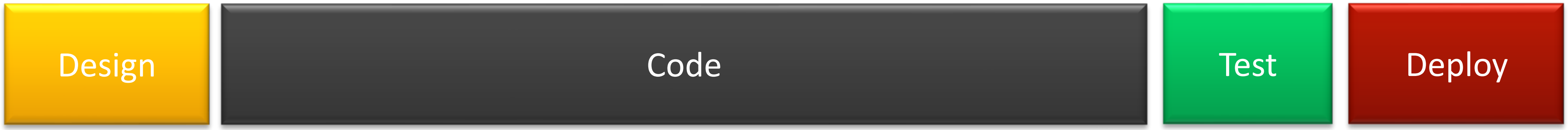
Example #4 – DevOps

- A mid-sized e-commerce company struggles with slow software releases and frequent production outages. Their traditional development and operations teams work in silos: developers finish coding and “throw it over the wall” to operations for deployment and maintenance.
- The company aims to accelerate release cycles, improve collaboration, and increase system stability. They decide to implement a **DevOps culture and practices** including continuous integration/continuous delivery (CI/CD), infrastructure as code (IaC), automated testing, and monitoring.

- Why DevOps is chosen?
  - Need for faster, more reliable software delivery
  - Desire to break down silos and improve communication
  - Reduce manual errors and deployment risks
  - Improve scalability and infrastructure management

Waterfall vs Agile vs DevOps

- **Waterfall**



- **Agile**

- **DevOps**

Why do DevOps matter in PM?

- DevOps matters in project management because it significantly enhances the efficiency, reliability, and scalability of software projects by bridging the gap between development and operations.

- Benefits:
  - Accelerated Delivery & Innovation
  - Automation & Efficiency
  - Improved Cross-Functional Collaboration
  - Reliability & Consistency
  - Better Monitoring & Quality Control
  - Governance, Security & Compliance
  - Support for Agile & Continuous Practices

Beyond DevOps: Emerging Cultures in Modern IT

These new cultures extend DevOps principles into specific domains:

- **DevSecOps:** Adds security early in the development process.
- **DataOps:** Helps teams manage and move data quickly and reliably.
- **MLOps:** Makes it easier to build, test, and run ML models.
- **AIOps:** Uses AI to find and fix IT problems automatically.
- **GreenOps:** Focuses on making IT systems more energy-efficient.
- **TestOps:** Includes testing in every step to catch bugs early.
- **GitOps:** Uses Git to safely automate software deployments.
- **FinOps:** Tracks and controls cloud spending with finance teams.

A quick 5 min stretch

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Requirement Gathering and Project Initiation

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Requirement Gathering

- Requirement Gathering is the process of identifying, collecting, and documenting the needs and expectations of stakeholders for a new or updated system or product.

- Why is it important?
  - Ensures the final product meets stakeholder needs
  - Reduces scope creep and rework
  - Forms the foundation for design, development, and testing
  - Facilitates better project planning and estimation

Requirement Gathering Techniques

- Interviews - One on One with stakeholders
- Surveys/Questionnaires - To father wide input quickly
- Workshops - Collaborative group sessions
- Observation - Watch users interact with current systems
- Document Analysis - Review existing system/process documentations

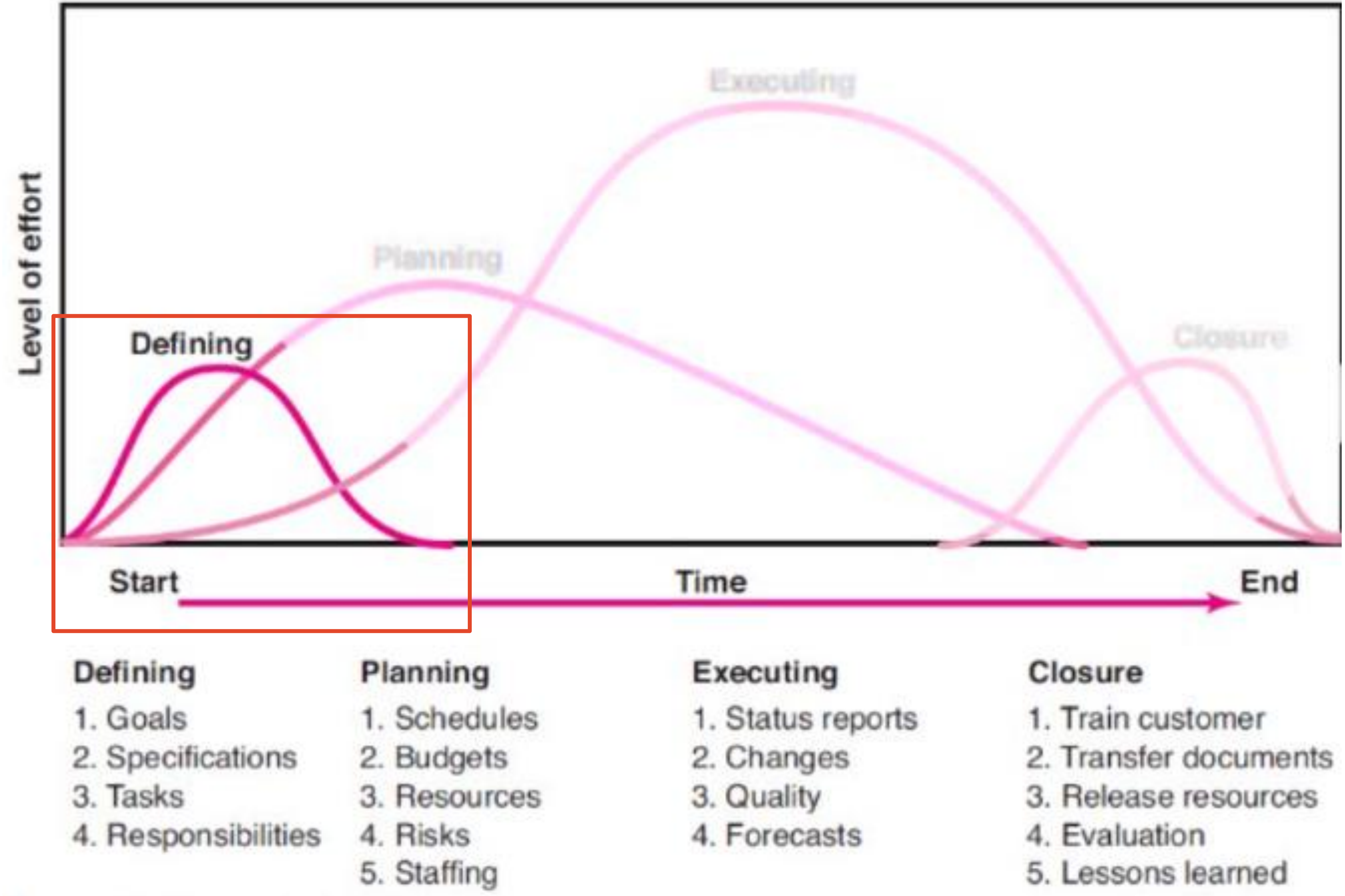

Figure 2.1: Project Life Cycle

Knowledge Area	Defining
Integration Management	Project Charter
Scope Management	Initial Scope Idea
Time Management	X
Cost Management	X
Quality Management	X
Resource Management	X
Communication Management	Identifying Stakeholders
Risk Management	Identifying High level risks
Procurement Management	X
Stakeholder Engagement	Identifying Stakeholders

Will be revealed gradually.

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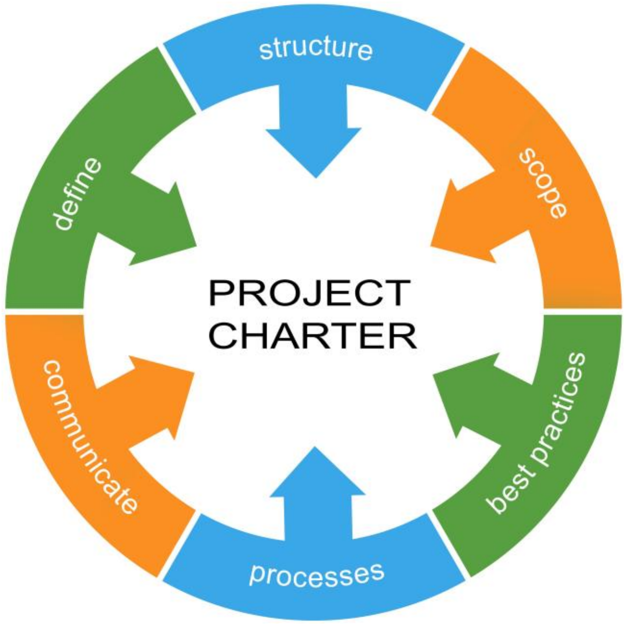
Introduction to Project Initiation/Defining

- Purpose:
  - Clarity on what is being done and why
  - Buy-in from stakeholders and leadership
  - A baseline for planning, execution, and success measurement
- Key activities
  - Identify the business need or problem
  - Define project goals and objectives
  - Defining the high-level scope
  - Identifying stakeholders
  - Appointing PM and key roles
  - Developing the Project Charter
  - Determining high level budget and timeline
  - Conduct initial risk and feasibility analysis
  - Align with Organizational strategy

Developing the Project Charter

- It is a formal document that authorizes the project and grants the project manager the authority to use organizational resources.
- Purpose:
  - Establishes a clear vision of the project
  - Aligns stakeholders and secures executive support
  - Provides a high-level roadmap for planning
  - Acts as a reference point throughout the project
- Tools used:
  - Expert Judgment
  - Business Case
  - Enterprise Environmental Factors
  - Organizational Process Assets



(Good, 2025)

An example of a Project Charter

PROJECT CHARTER

Project Title	Mapping of Mineral Resources on the sea floors using AI-powered visualization		
Project Start Date	August 01, 2022	**Project End Date**	August 6, 2024

Brief Background

Human technology has reached new heights in the past 20 years and is continuously reaching new peaks. Infrastructure, physical or technological, requires resources, and rapid development has started to affect the number of resources available for exploitation on the surface. This is leading us towards the tragedy of the commons, an economic problem where the resources would be highly sought after and exploited at such a rate that it would lead to the exhaustion of the resources and leave everyone in a lurch.

We would need to either find new alternatives or search for natural resources. In either case, the existence of natural resources is unquestionable. At this point in time, we can find our focus shifting towards seabed mining. Of the 360 million km^2 area of the seabed on the earth's surface, only a combined area of 1.4 million km^2 has been explored for mineral resources. This project would be aimed at creating the technology which would enable the mapping of the untapped resources in the ocean and provide data through which subsequent research can be conducted to mine the resources.

Project Objectives

- Cost-efficient and innovative Machine learning algorithm to detect deep-sea resources.
- Algorithm provides accurate interpretive maps of natural resource-filled regions in sea-beds.
- 3-D visualization of the natural resource deposits to aid project planning of natural resource collection.

Project Success Criteria

- Delivery of the completed software and hardware.
- Delivery of accurate 3-D Visualizations of natural resource deposits.

Project Milestones

- September 9, 2022 – Completion of Market Research
- October 21, 2022 – Completion of Project Plan
- June 22, 2023 – Finalization of Documentation
- March 28, 2024 – Finalization of Software
- March 20, 2024 – Procurement of Hardware
- May 09, 2024 – Complete integration of Hardware and Software
- August 5, 2024 – Finalization of product
- August 6, 2024 – Product Release

Budget Information

- Total cost - est. AUD 4.00 million

ROLES AND RESPONSIBILITIES

Name	Role	Responsibility
	CEO	Project Sponsor, monitor the project
	IT department Head	Management of technical operations
	Project Manager	Plan and execute project whilst coordinating multiple teams
	Team Director	Identify team goals and coach team members and links the Project manager with the team
	Finance Officer	Manage the budget, Prepare financial reports
	Market Research Analyst	Research, compile and analyze information on products

SIGNATURES OF STAKEHOLDERS

Requirement Analysis Across Different Methodologies

- **Waterfall**
  - Requirements gathered **fully upfront**
  - Detailed documentation
  - Formal approval before development
  - Changes are costly and controlled

- **Agile / Scrum**
  - High-level requirements at start
  - Detailed requirements per sprint
  - User stories + acceptance criteria
  - Continuous stakeholder feedback

Requirement Analysis Across Different Methodologies contd.

- **Kanban**
  - Requirements added continuously
  - Pulled based on team capacity
  - Flexible prioritization
  - Less formal documentation

- **DevOps**
  - Requirements evolve from monitoring & feedback
  - Strong focus on automation & deployment needs
  - Continuous improvement loop

Key Differences in Requirement Approach

Methodology	When Defined	Documentation Level	Change Flexibility
Waterfall	Beginning	Heavy	Low
Agile	Iteratively	Moderate	High
Kanban	Continuous	Light	Very High
DevOps	Continuous + Feedback	Light to Moderate	Very High

Stakeholder Identification and Engagement

- A stakeholder is anyone who impacts or is impacted by the project: positively or negatively.

- Examples: customers, sponsors, team members, suppliers, regulators, etc.

- Why is this step important?
  - Ensures early buy-in and support
  - Helps manage expectations and influence
  - Reduces resistance and improves communication
  - Aligns project goals with stakeholder interests

Stakeholder Identification and Engagement contd.

- Key activities involved:
  - **Identify Stakeholders:** Use brainstorming, expert judgment, documents, etc.
  - **Analyze Stakeholders:** Determine power, interest, and influence
  - **Prioritize:** Categorize using tools like the Power/Interest Grid
  - **Engage & Communicate:** Tailor strategies based on stakeholder type
  - **Monitor Engagement:** Adjust plans as stakeholder needs change

Tools and Techniques used in Defining stage

- Expert Judgement
- Brainstorming and Workshops
- SWOT Analysis
- Stakeholder Analysis
- Meetings and Interviews

Stakeholder Engagement Across Methodologies

Stakeholders are **identified early in all project methodologies.**
However, the **style and frequency of engagement differ**.

Aspect	Waterfall	Agile / Scrum	Kanban	DevOps
When Identified	Early, during initiation	Early + continuous refinement	Continuous	Continuous
Level of Formality	High (documents, approvals)	Moderate	Low to Moderate	Moderate
Stakeholder Involvement	Limited after requirements phase	Frequent (Sprint Reviews)	Ongoing feedback	Continuous + operational feedback
Communication Style	Structured, scheduled	Collaborative, iterative	Flexible, on-demand	Cross-functional, integrated
Change Handling	Formal change requests	Adapted each sprint	Anytime	Continuous improvement

Key Differences & Managerial Insight

- **Waterfall**
  - Structured, milestone-based engagement with formal approvals.
  - Heavy involvement during requirements and final acceptance; limited interaction during development.
- **Agile / Scrum**
  - Continuous collaboration and regular stakeholder feedback.
  - Frequent reviews (e.g., Sprint Reviews) allow ongoing refinement of requirements and priorities.
- **Kanban**
  - Flexible, flow-based engagement.
  - Stakeholder input can be incorporated anytime, depending on work demand and priorities.
- **DevOps**
  - Cross-functional engagement extending into operations.
  - Stakeholders remain involved even after deployment through monitoring and continuous improvement.

Common challenges in Project Initiation/Definition

- Even before planning begins, projects can stumble. These are the most frequent issues during initiation:
  - Unclear objectives and scope
  - Lack of executive support
  - Poor Stakeholder engagement
  - Incomplete or Weak business case
  - Unrealistic timelines and budgets
  - Undefined roles and responsibilities
  - Inadequate risk awareness

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For You!

End of Lecture Questions

- What is the main purpose of the project initiation phase?
- Name three key elements that are included in a Project Charter.
- Why is stakeholder identification important during initiation?
- How do project methodologies influence project initiation?
- Describe the Power/Interest Grid and how it's used.

Case Study

- You are assigned as the project manager for a mid-sized company looking to launch a new e-commerce platform. The CEO wants the project to be completed within 6 months but has not yet defined detailed requirements. The marketing team is eager to start, but the IT department is unsure about platform feasibility. There is no formal project charter yet.

  - Identify three immediate steps you would take as a project manager to define this project.
  - What key stakeholders can you identify in this case, and how would you prioritize them using the Power/Interest Grid?
  - What challenges in project initiation are evident in this scenario? How would you address them?
  - What risks can you foresee at this stage, and how would you begin assessing them?

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